

The Structure of Autobiographical Memory and the Event History Calendar: Potential Improvements in the Quality of Retrospective Reports in Surveys

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This paper reviews the literature on the structure of autobiographical memory and points to the use of event history calendars as a survey methodology that reflects this structure. Autobiographical memory structure is characterised as an hierarchical network that includes extended, summarised, and specific events, and that permits retrieval of past events through multiple pathways that work top-down in the hierarchy, sequentially within life themes that unify extended events, and in parallel across life themes that involve contemporaneous and sequential events. Traditional survey questions tend to segment related aspects of autobiographical events from one another, and do not reflect the interrelatedness of events as indicated within the structure of autobiographical memory. In contrast, event history calendars do promote sequential and parallel retrieval within the autobiographical memory network. By reflecting the structure of autobiographical memory, the use of event history calendars has considerable potential in assisting respondents to reconstruct their personal pasts more completely and accurately, maximising the quality of retrospective reports.

INTRODUCTION

The quality of survey research is fundamentally dependent on the validity of respondent reports, including the ability of respondents to accurately report on their past behaviours, or events that they have witnessed. Survey researchers have been aware that psychological research on autobiographical memory—that is, memory of one's personal past—is relevant to their goals of understanding errors that have been observed in their respondents' retrospective reports (e.g. Schwarz & Sudman, 1994). Errors have been found in people's reporting of what, when, and how often events have happened (Jobe, Tourangeau, & Smith, 1993). Based on understanding the nature of autobiographical memory, explanations for these errors have been sought with the ultimate goal of improving the accuracy of survey report.

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The Structure Of Autobiographical Memory
Memory

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Illustrating limitations in autobiographical recall, survey reports have revealed gross underreporting of even distinctive events such as hospitalisations (Cannell, Fisher, & Bakker, 1965), motor vehicle accident injuries (Cash & Moss, 1972), and periods of unemployment (Mathiowetz, 1986). Length of retention interval and duration of event have been shown as factors affecting underreporting. For example, Cannell et al. found that respondents failed to report only 3% of hospitalisations when asked within 1 to 10 weeks of the event, but failed to report 42% of hospitalisations when asked 1 year after the event; Cannell et al. also found that underreporting occurred most often for short-duration hospital stays (1–4 days). With respect to remembering when events happened, respondents tend to report events as having occurred more recently than in actuality, a phenomenon known as *forward telescoping* (e.g. Bachman & O'Malley, 1981; Thompson, Skowronski, & Lee, 1988). In explaining telescoping, researchers have suggested processes that involve either biases in temporal representation (Bradburn, Rips, & Shevell, 1987; Brown, Rips, & Shevell, 1985) or artifacts of being asked to report within boundaries that are imposed by asking temporal reference period questions (Huttenlocher, Hedges, & Bradburn, 1990; Rubin & Baddeley, 1989). Finally, accuracy in reporting how often events happened has been found to be a complex interplay of such variables as the frequency, similarity, and regularity of events, the length of the retention interval, the use of closed or open-ended questions, and whether respondents engage in enumeration or estimation processes or some combination of these retrieval strategies (Brown, 1995; Burton & Blair, 1991; Menon, 1993; Menon, Raghubir, & Schwarz, 1995).

Although such work has been informative, it has largely ignored psychological studies that have focused on the content of autobiographical knowledge and how such knowledge is organised and structured (Barsalou, 1988; Conway, 1996; Linton, 1986). This psychological work has concentrated on the invariances that are found in the remembering process as people reconstruct their personal past. These invariances specify that certain types of autobiographical memories are thematically and temporally structured within an hierarchical ordering. Further, retrieval of episodic information concerning what has happened in the past has been shown to be facilitated by using memories from higher-order levels as cues (Conway & Bekerian, 1987). Although any particular retrieval usually follows one particular pathway through the autobiographical memory hierarchy, the richness of cues that are available within the autobiographical memory structure potentiates the ability to increase the quality and completeness of retrospective reports, by encouraging the activation of multiple pathways at one time.

This paper will illustrate that our present understanding of the structure of autobiographical knowledge can be used to improve the quality of survey respondents' retrospective reports of what, when, and how often events happened. In illustrating this potential, there will be special concentration on

an already developed technique in survey methodology known as the event history calendar (Caspi et al., 1996; Freedman et al., 1988; Kessler & Wethington, 1991). Although definitive studies have yet to be conducted that directly contrast the effectiveness of event history calendars in comparison to standard survey questions, the use of event history calendars has been shown to lead to high-quality retrospective reports even after retention intervals of several years. Remarkably, utilisation of event history calendars relies on the use of both thematic and temporal information which has been shown to be so vital as an aspect of the structure of autobiographical knowledge. Thus, the success of the event history calendar is largely traceable to the use of many facets of autobiographical memory structure to improve the quality of retrospective reports. In addition to discussing the event history calendar, I will devote attention to how this methodology can potentially be improved by an even greater reliance on temporal and thematic cues available within the organisation of autobiographical knowledge.

This paper is divided into two major sections. The first section will concentrate on what is known about the structure of autobiographical knowledge. The second section will illustrate the implications of this structure towards improving the quality of retrospective reports in surveys.

AUTOBIOGRAPHICAL MEMORY STRUCTURE

Autobiographical memory may be best described as involving the remembrances of events that have been personally experienced (Neisser, 1986). Personally experienced events are structured in autobiographical memory by hierarchically ordered *types of memories* for events that vary in their scope and specificity, and this structure is organised along *temporal and thematic pathways* which guide the retrieval process.

Types of Memories

Although there have been slightly divergent views regarding the various types of memories that people recover when remembering their personal past (e.g. Barsalou, 1988; Brewer, 1986; Conway, 1996; Linton, 1986), a summary of this literature reveals three main types. The most general types of memory, which comprises the top and middle of the hierarchy, is for *extended events*. Memories for extended events reveal the temporal nature of autobiographical memory, as such events are extended in time for periods as long as many years to as short as just a few days. Comprising the middle of the hierarchy are memories for *summarised events*, which emphasise the thematic aspect of autobiographical memory. In remembering summarised events, individuals are considering the common themes that underlie events of the same kind. Finally, the most detailed type of memory at the bottom of the hierarchy is for *specific events*. Memories of specific events include the perceptual and episodic information that provides

a sense of 'reliving' a particular episode as it originally occurred. As the remembering of a specific event involves reconstructing details from a single event, the time-span of these events is of short duration, from many seconds up to and including several hours (Conway, 1996).

Extended Events. At the top of the hierarchy, extended events comprise the most fundamental organisational structures, which nest all other types of autobiographical memories. Extended events may span many years (e.g. "When I work at X") or be fairly short in duration, perhaps extending for just a few days (e.g. "trip to Washington, DC"), and they have identifiable beginnings and endings (if not continuing into the present), although the exact endpoints may be fuzzy rather than exact. In addition, extended events are often characterised by being discontinuous in their flow, in which other activities are interspersed (Barsalou, 1988). For example, one doesn't work at "X" for 24 hours a day. Additionally, extended events are nested within one another (Barsalou, 1988; Neisser, 1986), such that a larger extended event (e.g. "first job as lawyer") contains other smaller extended events (e.g. "period of training"). Moreover, any particular extended event consists of a temporal ordering of those elements that it contains, such as the extended event of "second marriage" consisting of a temporal sequence of smaller extended events such as "falling in love", "living together", and "getting married".

According to some theorists (Conway, 1996; Conway & Rubin, 1993; Linton, 1986; Schooler & Herrmann, 1992), certain extended events serve as the primary organisational units to autobiographical knowledge and are also qualitatively different from other extended events. These more primary extended events, which will be referred to as *lifetime periods* (Conway, 1996), represent major thematic divisions in one's autobiography that are particularly noteworthy in determining one's concept of self (e.g. wife, psychologist, son, student, etc.), and the changes in this concept of who one is across the lifespan (see Fig. 1 for an illustration of Conway's model). As extended events, lifetime periods encompass a certain bounded period of time (e.g. "when I lived with 'X'"), and nest other events to which they index. Lifetime periods are also related by sharing higher-order themes such as a relationships theme and a work theme (Conway, 1996; Linton, 1986). Within a theme, the lifetime periods are sequentially ordered by time, such that "when I lived with 'X'" (within relationships theme) precedes "when I was married to 'Y'". Across themes, more than one lifetime period may overlap during the same period of time, depending on which higher-order themes are governing one's sense of self at that moment in time. For example, there may be overlap of being a father during "when I was married to 'Y'" with being a cognitive psychologist during "when I worked at 'Z'". Moreover, lifetime periods, and especially the themes of which they are a part, are not only based on actual experience, but are also governed by shared cultural norms

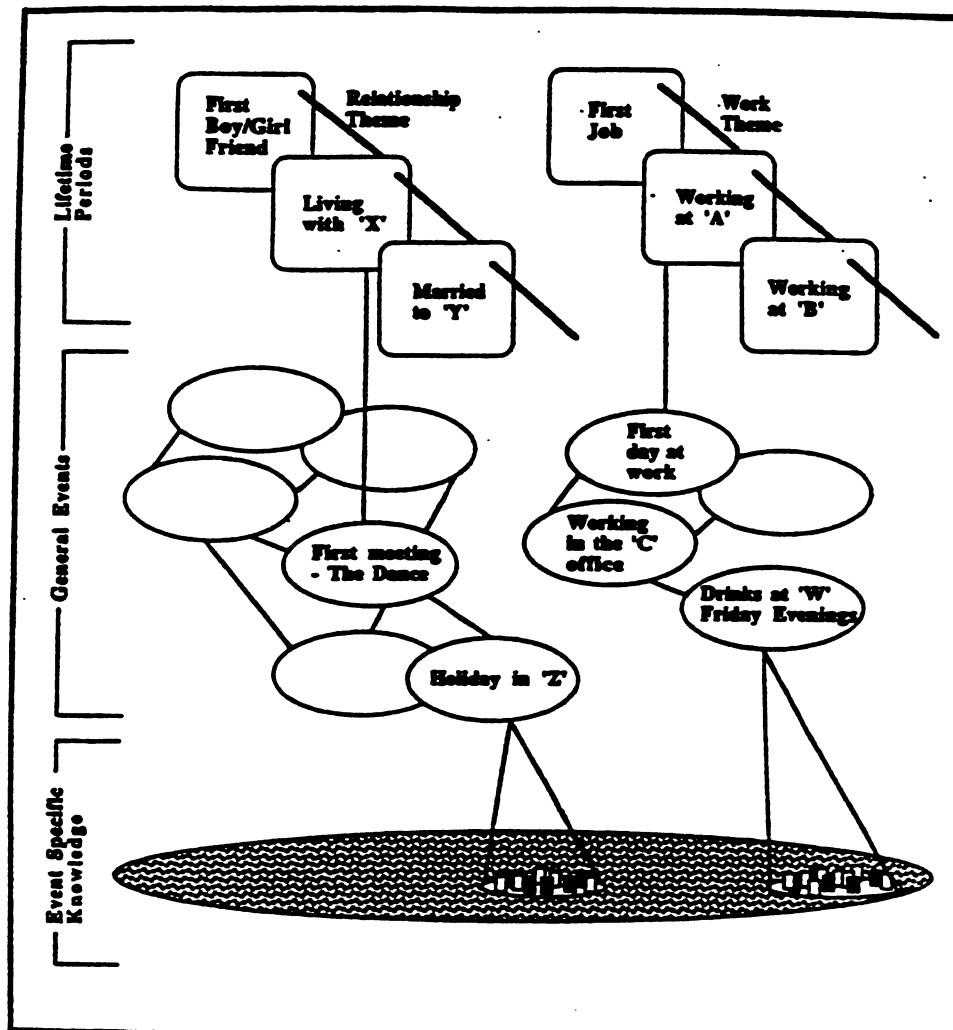


FIG. 1. An illustration of Conway's (1996) model of hierarchical autobiographical knowledge structures. Copyright ©1996 by Cambridge University Press. Reprinted with permission.

that largely determine the ways in which the events of life will be experienced and interpreted.

Summarised Events. A memory for a summarised event refers to a class of events of the same kind, reflected in phrases such as "I watched TV a lot" or "I usually ate lunch with my dad" Barsalou (1988) found that most of the statements in the verbal reports of people who were stopped on the campus of Emory University during an autumn semester (and who agreed to relate their experiences from the past summer) were in the form of summarised events. The

prominence of summarised events in these reports suggests their important role in being able to efficiently communicate to others (and perhaps to efficiently remember oneself) the specifics of what happened during one's personal past. In Conway's (Conway, 1992, 1996; Conway & Rubin, 1993) model (see Fig. 1), summarised events together with the shorter-term extended events (which are not lifetime periods) belong to a class of memories depicting *general events*, which are considered a *basic level* of representation (see Rosch, 1978, regarding basic level for categorisation). As a basic level of representation, summarised events are not too general, which would lead to an inability to convey anything specific about the past, nor too specific, in which the flow of past events could not be adequately expressed.

Summarised events are abstractions of the typical aspects of similar events, and thus, by themselves, primarily represent thematic knowledge about recurring temporal patterns of experience. Linton (1982) discussed a summarised event as a semantic memory that forms in the repetition of two or more similar events, with the episodic information for any one event becoming lost and only the generalised information from the semantic memory being retained. Neisser (1981) introduced the term *repisodic memory* to refer to the representation of repeated episodes, with the idea that remembered details may reflect information in repisodic memory and not be veridical concerning any specific event (although the rememberer believes in their veridicality). In the wider picture, the temporal organisation of extended events is permeated to summarised events, as summarised events are nested within extended events. For example, an undergraduate's summer may be represented as an extended event that nests smaller extended events such as an early summer job, a period of vacationing, and a later period of staying at home with one's parents; each of the smaller extended events nest representations of summarised events such as lunches with co-workers during the job and evening hikes during the vacation (Barsalou, 1988, p.225).

Specific events. At the bottom of the hierarchy are memories for specific events. Richness of autobiographical memory is gained in the remembering of specific events, which may contain perceptual and contextual information that provides a sense of reliving a past event. Extended in time, specific events, such as "time I met Angela", "saw a play", or "filled out an application", are best characterised as being continuous in their time course and spanning minutes up to many hours. Within the hierarchy, specific events may be indexed by extended and/or summarised events.

In Conway's (1992, 1996) model (see Fig. 1), the perceptual and contextual information (termed as *event-specific knowledge*) of specific events exists as an undifferentiated pool of images and details. These details become organised during the reconstructive remembering process by attaching to the thematic and temporal knowledge structures of higher-order events. In this view, any

particular episodic detail may be reconstructed within the bounds of more than one specific event.

Anderson and Conway (1993) explored the organisation of event-specific details within specific autobiographical memories, such as remembering the "time I met Angela". They discovered that certain distinctive details (e.g. "dancing with Angela") encouraged the recognition of a specific memory as a whole, whereas a greater number of details could be remembered when people recalled events in forward chronological order. These results suggest that thematic and temporal information will structure specific episodes in much the same way as they are used to structure the higher-order extended and summarised events. As for remembering specific events, the distinctive details are indices to specific memories from higher-order structures, whereas once indexed, the chronological ordering of details aids in remembering more of what had happened.

Thematic and Temporal Pathways

As we have seen in the previous section outlining types of autobiographical memories, thematic and temporal knowledge serve as the basic organising principles that underlie all levels of autobiographical memory structures. For example, extended and summarised events are nested within higher-order extended events both thematically (in reference to particular persons and types of events) and temporally (in reference to being placed within a particular point in time on the basis of life roles and the sequence of events). Taken as a whole, the temporal and thematic organisation of autobiographical knowledge serves as points of association among these various memory types within a rich network of retrieval pathways that comprise this knowledge. In any particular retrieval, however, only some of the available pathways are likely to be used.

In describing the retrieval of autobiographical memories, researchers have highlighted one process as being top-down in which the thematic and temporal information of higher-order structures cue the remembering of more specific information (Barsalou, 1988; Conway & Bekerian, 1987). In addition, however, people navigate through autobiographical knowledge in other ways, such as when they order a sequence of extended events, or when they retrieve from parallel events (Barsalou, 1988).

Top-down Retrieval. In Conway's model (Anderson & Conway, 1993; Conway, 1996; Conway & Bekerian, 1987; Conway & Rubin, 1993), lifetime periods hold the key in containing various types of thematic and temporal knowledge concerning such things as people, aspirations, and significant events. Such thematic and temporal knowledge, in a broad sense, serves to index many general events that are associated with particular lifetime periods. For example, "living with 'X'" contains thematic and temporal information about the

relationship with 'X' which will index the memory for "holiday in Italy with 'X' ". In addition, general events contain thematic and temporal information that indexes memories of more specific events, such as "dinner with 'X' on our last night in Rome". Further, this model considers this hierarchical organisation as consisting of *autobiographical memory organization packets* (A-MOPs) that are tied to the thematic contents of specific life periods. Thus, although one lifetime period ("living with 'X' ") can be contemporaneous with another lifetime period ("when I worked at 'Y' "), the thematic contents of these respective lifetime periods would consist of separate A-MOPS that would index their own kinds of life events. Evidence that favours top-down retrieval within A-MOPS is provided by response-time studies and in the analyses of the protocols of subjects retrieving specific events.

In one response-time study, Conway and Bekerian (1987) found faster response times in the retrieval of specific events, which were cued by general events, in a condition in which lifetime periods were used as primes in comparison to a no-prime condition. Further, they also found that these higher-order autobiographical memory primes and cues speeded retrieval in comparison to other types of primes and cues that were based on semantic categories or activities. These results support the hypothesis that higher-order autobiographical memories facilitate the retrieval of specific events, as predicted by top-down retrieval models.

In another response-time study, Brown (1990; Brown, Shevell, & Rips, 1986) demonstrated that the response times for political and nonpolitical public events was dependent on which contemporaneous periods people had in mind. Subjects were asked to sort public events either as occurring in one of two US Presidential administrations (Carter or Reagan), or in one of two lifetime periods (student in high school or college). Subjects were selected such that they were in high school during the Carter administration, and in college during the Reagan administration. Brown found that political events were sorted more quickly into Presidential administration periods than they were sorted into lifetime periods, but that nonpolitical events were sorted more quickly into lifetime than administration periods. Although addressing more directly the relationship between public and personal events rather than the structure of autobiographical memory *per se*, Brown's results are suggestive that lifetime periods index different aspects of the autobiographical memory base, consistent with the model that autobiographical knowledge consists of A-MOPS.

Verbal protocol analyses are also supportive of top-down retrieval. In one study, Conway (1996, see Conway's figure 3.3) presented subjects with cue words (e.g. chair, restaurant) with explicit instructions to use them in retrieving specific events. Protocols revealed subjects working through lifetime periods and general events before reporting the details of specific events. Similar results were found with subjects aged in their late 20s or early 30s who were asked to

report details of specific events that occurred while they were 18 years old (Conway & Bekerian, 1987, p.131).

Sequencing and Parallel Retrieval. In addition to top-down retrieval, there are retrieval strategies that involve the chronological sequencing of events within and across themes. These retrieval strategies appear to become apparent when individuals are asked to report on their autobiographical past for an extended period of time, such as in Barsalou's (1988) study, when subjects were asked to report about the past summer. The top-down processing observed by Conway (1992, 1996; Conway & Bekerian, 1987) may be more apparent when asking about single specific episodes. Thus, the nature of the request probably affects the retrieval strategies that are employed.

Retrievals governed by the recall of chronological events complement Conway's (1992, 1996; Conway & Bekerian, 1987) emphasis that thematic content governs autobiographical retrieval. In Barsalou's (1988) study, although subjects were instructed to recall events on the basis of what first comes to mind, most people started their reports at the beginning of the summer, working chronologically through extended events that occurred at the first part of the summer. As Barsalou's subjects were likely to be associated with Emory University (they were recruited on campus), the boundary between spring and summer would be a particularly notable transition in daily activities which would coincide with the beginning of summer (and the end of the regular school term). As found in other work, which has shown that events that occur at transition points are more memorable than other events (Pillemer, Goldsmith, Panter, & White, 1988; Robinson, 1986), subjects beginning their reports at the start of summer involves the use of thematic information about changes in activities that are associated to a temporal location. In a similar vein, Means, Nigam, Zarrow, Loftus, and Donaldson (1989) found that subjects' free recalls of health events for the past year often included the chronological ordering of events that pertained to the same health condition. Thus, both the thematic information of type of condition and the chronological ordering of events were used in remembering these past events.

The temporal sequencing of extended events as observed by Barsalou (1988) indicates the predominance of a horizontal rather than top-down process. Yet, as there often was an intermixing of summarised and specific events in subjects' reports, top-down processing was also occurring in these retrievals. In addition to sequencing, Barsalou observed that subjects at times followed *parallel tracks of events* in which one chronology of extended events dealing with one theme (e.g. project at school) was followed in the protocol by a second chronology that was contemporaneous with the first but dealt with a different theme (e.g. activities with family members). Similarly, Means et al. (1989) found that work-related events were at times used as junctures to remember health events. This jumping across thematic tracks suggests an interrelationship among thematically

different extended events that is not captured by the notion of A-MOPs, which, in comparison, considers extended events as being largely thematically encapsulated.

Barsalou (1988) contends that the extended events both within and across themes are interrelated by the *logic of goal attainment* (see Fig. 2 for an illustration). For example, within a work theme, the activities of one's first job (e.g. Assistant Professor at 'X') may enable one to develop skills for a second job (e.g. Associate Professor at 'Y'). Across themes, one's first job (work theme) may be where one meets one's spouse (relationship theme), or losing one's job may contribute to an eventual divorce. Thus, the logic of goal attainment is based on the aspirations of the self, and successes and setbacks in meeting these

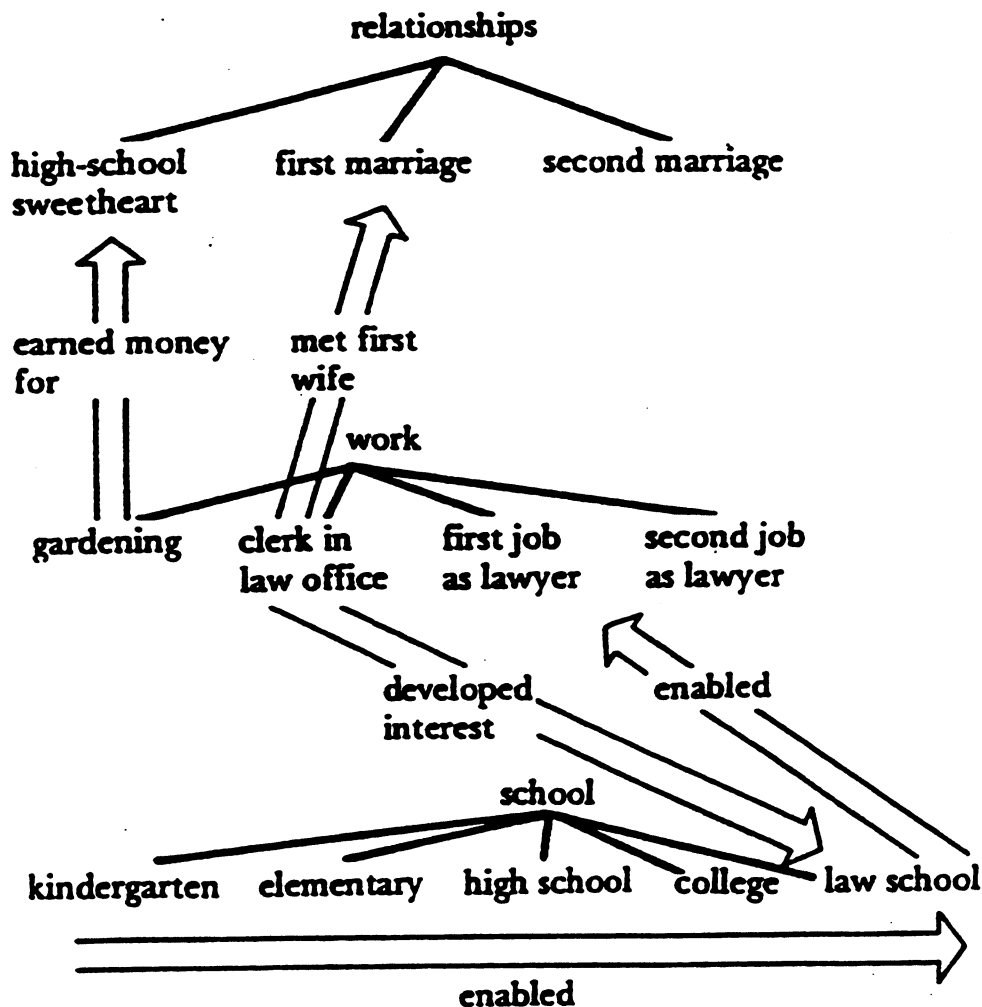


FIG. 2. An illustration of Barsalou's (1988) notion that logic of goal attainment structures interrelate events both within and across extended events. Copyright ©1988 by Cambridge University Press. Reprinted with permission.

aspirations. The importance of goal attainment in autobiographical memory is shared by Conway (1992, 1996; Conway & Rubin, 1993) who argues that autobiographical themes emerge from changes to the self-concept and aspirations and fears that surround conceptions of possible selves (cf. Markus & Nuriur, 1986). In a vivid demonstration of goal attainment, Robinson (1992) finds that verbal reports of first experience memories, such as the first time one drives a car or the first romantic relationship, are filled with descriptions of how these events had shaped the conception of self.

In combination with top-down retrieval, the notions of sequencing and parallel retrieval outline the existence of numerous interconnected thematic and temporal pathways that can be used to remember specific autobiographical events. Although only a small portion of available pathways will be used in any one retrieval, encouraging individuals to use the cues that are available in multiple, interconnected pathways increases the potential of maximising the quality and completeness of remembered events. This potential of improving the quality of autobiographical remembering, through the encouragement of respondents to examine more fully the interconnections that exist within their autobiographical networks, has yet to be fully appreciated by survey researchers. The next section will address this issue more fully.

AUTOBIOGRAPHICAL MEMORY STRUCTURE AND SURVEYS

Traditional standardised survey interviews that ask for retrospective reports consist of a set of discrete questions, usually about specific events, that are separated by topic. The University of Michigan's *Panel Study of Income Dynamics*, a survey of family economics, initially asks questions about housing (e.g. "Have you moved any time since the spring of 1996? ... What month and year was that?"), then switches to employment (e.g. "In what month and year did you start working in your present position?"), and then asks questions about time away from work (e.g. "Did you miss any work in 1996 because you were sick? ... How much work did you miss? ... When was that? ..."). As another example, the University of Michigan's *Health Field Study*, a survey based on the *National Health Interview Survey*, at one point asks about hospitalisations (e.g. "Since July 1, 1996, have you been a patient in a hospital overnight? ... How many different times did you stay in any hospital overnight or longer since July 1, 1996? ..."), then about health visits (e.g. "How many times have you gone to a hospital emergency room about your own health since July 1, 1996? ... How many times have you seen or talked to a medical doctor or assistant about your own health at a doctor's office or at an HMO since July 1, 1996? ..."), then about specific health services (e.g. "Have you had a tetanus shot during the past ten years?"), and then about injuries ("How many injuries, accidents, or poisonings have you had since July 1st, 1996 that required you to be seen by a medical doctor or assistant?").

In terms of the structure of autobiographical memory, and more specifically, with regard to the number of cues that can be made available by encouraging multiple retrieval routes, traditional survey questioning methods are found wanting. On the positive side, survey questionnaires generally do encourage top-down retrieval. As survey researchers are aware of the need to order questions in ways that facilitate smooth transitions and optimise flow (Sudman & Bradburn, 1982), in most cases the general context is introduced before more specific questions are asked. More negatively, however, there is little or no encouragement for sequencing and parallel retrieval. Sequencing is encouraged in situations that ask for the remembering of extended events, and although survey questions at times do appear interested in such events (note the questions on time off from work and hospitalisations quoted earlier), their intent appears to focus more on discrete points in time rather than on a continuous stream of events. Worse still is the total lack of encouragement for parallel retrieval, that is, the remembering of interconnected aspects of one's personal past that associate themes to one another. As survey questionnaires typically move from topic to topic, respondents are encouraged to segment their paths into units that are largely unrelated to one another. Taken together, the lack of use of cues available in the structure of autobiographical memory may be largely responsible for the errors that have been seen in respondents' attempts to report what, when, and how often events have happened.

Event History Calendars

A different approach to asking survey questions involves the use of event history calendars, which have considerable potential in utilising the top-down, sequential, and parallel cues that are available in the structure of autobiographical memory. In contrast to traditional survey questions that emphasise events as discrete occurrences, with the event history calendar, respondents are encouraged to consider various events that constitute their personal pasts as contained within broader thematic streams of events. Not only can respondents note the interrelationship of events within the same themes (top-down and sequential retrieval) but, depending on which themes are represented by the calendar, respondents can also note the interrelationships among events that exist within different themes (parallel retrieval).

Pioneered by researchers interested in life histories (e.g. Freedman et al., 1988), the use of event history calendars has been based on the notion that life histories consist of *trajectories* and *transitions* (Elder, 1985). Life trajectories are long-term extended patterns that reflect major domains such as work, living arrangements, and relationships, and as such are conceptually similar to *themes* as discussed in the literature on autobiographical memory. Transitions are embedded within trajectories and represent points in time in which abrupt changes occur, such as getting married or divorced, or finding that first job.

Transitions are therefore beginning and endpoints of *lifetime periods* or other significant extended events. In addition, the dynamic aspect of the life course evolves as an interplay among trajectories and their transitions, which are interdependent on one another. Again, this interdependence is reflected in autobiographical memory as the interrelationship among thematically different extended events.

Event history calendars, then, were first developed as attempts to assist respondents in recalling their dynamic pasts. Consider the life history calendar of Freedman et al. (1988). As depicted in Fig. 3, this calendar presented five themes that examined places of residence, persons residing with and their relationships, schooling, work, and military service. Each theme was represented by its own relevant timelines that covered a 10-year period. Respondents were asked to report the month and year of all events including certain specific events (e.g. births of offspring) and the beginning and endpoints of extended events (e.g. marriages, cohabitations). Superficially, the life history calendar mirrors the kinds of information that will be available in the remembering of lifetime periods, as it represents those themes that are likely to be shared by members of a culture. More importantly, the life history calendar encourages an extensive use of the autobiographical memory network and the multiple paths associated with top-down, sequential, and parallel retrieval strategies.

Top-down Retrieval. Organised by themes, the life history calendar encourages top-down retrieval. For example, the first timeline, dealing with places of residence, encourages respondents to report in what places (e.g. cities, towns) they have lived (and when), a task that for most respondents is done quite easily, perhaps because the focus is on retrieving lifetime periods. Another timeline dealing with marital events and cohabitation asks for reporting of the persons with which one is involved. Another example that required more detailed information involved the reporting of work histories, in which respondents were asked to report on the number of weekly hours worked. The encouragement of top-down retrieval with the work histories would have been enhanced if, in addition to weekly hours, respondents had been asked explicitly to report their lifetime periods regarding the places in which they were working. The lifetime periods could serve as a retrieval cue for remembering weekly work hours.

Sequencing and Parallel Retrieval. In comparison to traditional standardised survey techniques, the life history calendar excels in encouraging retrieval through cues obtained by navigating through sequential pathways within themes and parallel pathways across themes. As administered by Freedman et al. (1988), the life history calendar emphasised sequential retrieval, as interviewers were instructed to elicit the chronological retrieval of events within the first presented theme (places of residence), and then move on to other themes in the order of

		AGE		15												16																					
		YEAR		1976												1977																					
		MONTH		J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D				
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LIVED WITH WHOM	84	PARENTS TOGETHER		Y	N																																
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		DOMESTIC PARTNER		Y	N																																
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		OTHER																																			
SCHOOL	813	FULL-TIME																																			
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		NONE																																			
WEEKLY WORK HOURS	814	30 OR MORE																																			
		15-29																																			
		1-14																																			
		NONE																																			
MILITARY	815	IN SERVICE		Y	N																																
MONEY	816	PAID ROOM AND BOARD (YELLOW BOOKLET, R3)		Y	N																																
MONEY	817	EDUCATION		Y	N																																
		CAR		Y	N																																
		EMERGENCIES		Y	N																																
		RENT		Y	N																																
		GIFTS		Y	N																																
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823		ANY FINANCIAL HELP		☐																																	
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FIG. 3. The truncated and reduced reproduction of the life history calendar used in the Study of American Families and discussed by Freedman et al. (1988). Copyright ©1988 by the American Sociological Association. Reprinted with permission.

appearance on the calendar. Checks were made to ensure that no gaps occurred in the reporting of events. As for parallel retrieval, respondents may have spontaneously coordinated responses from those timelines that had already been answered to those responses given in the later stages of the interview. In face-to-face interviewing, the visual presentation of each theme along the same time dimensions facilitated examining parallel tracks of events. Memory researchers (Barsalou, 1988; Neisser, 1988) have noted that spatial perceptions and metaphors are used to represent time (e.g. "those days are behind me") and the calendar's spatial representation of time provided a clear means to examine those events that were reported to have occurred earlier, at the same time, and later than other events. Thus, for example, respondents who went to work part-time following the birth of a child could see whether they had appropriately coordinated these events in their reports. Additionally, interviewers, in examining reports for inconsistencies, could query respondents about the accuracy of across-theme reports that appeared out of the ordinary. For example, a respondent who reported both working and going to school full-time at the same time could be asked to clarify the accuracy of these responses.

Barsalou's (1988) notion regarding the *logic of goal attainment* captures the interrelationship of extended events both within and across themes due to the logical association of one extended event (e.g. birth and infancy of 'X') with other extended events that are either contemporaneous (e.g. working part-time at 'Y') or that occur later (e.g. 'X' attends 'Y' elementary school). Thus, the logic of goal attainment, available in both sequential and parallel retrieval, provides information regarding the relative temporal ordering of events. Reconstructive models of event dating are based on the notion that the contents of remembered experiences contain clues regarding when they had occurred (Friedman, 1993; Thompson, Skowronski, Larsen, & Betz, 1996). These reconstructive models are supported by evidence that greater precision in dating past events occurs for memories in which there exists a greater degree of content or memorability (Bartlett & Snelus, 1980; Brown et al., 1985; Thompson et al., 1988). The temporal sequencing of events associated by the logic of goal attainment can also serve as information that will facilitate the attainment of a more precise reconstruction of when events had happened. Additionally, the multiple cues generated in event history calendars can promote the remembering of otherwise forgotten events that will affect the accuracy and completeness in reporting what events had happened (Anderson & Pichert, 1978; Fisher & Quigley, 1992). Finally, the reporting of how often events had occurred transpires indirectly from respondents remembering what had happened across the recall period demarcated by the timelines, and in which the number of events per unit time can be established by the researcher after the data are collected. Gaining "how-often" information through the event history calendar contrasts with traditional survey questions, which directly ask for reports of the number of events

within a given reference period, and which often provide a dearth of cues to help respondents remember the occurrence of these events.

Response quality. Of course, arguments concerning the efficacy of event history calendars are vacuous unless there is evidence that their use leads to accurate reporting of past events. Freedman et al. (1988) were able to compare the retrospective reports of school attendance and weekly work hours in the life history calendar against a concurrent report gathered with standard interviewing that had taken place five years earlier. In examining the same month of the prior concurrent report, 87% of respondents gave identical answers to both interviews regarding full-time, part-time, or no school attendance. Respondents retrospectively reported themselves as being slightly busier in their employment status in comparison to the concurrent report. Identical answers regarding working full-time, part-time, or not working were provided by 72% of respondents, which improved to 83% when collapsing full- and part-time work. In another application of a life history calendar, Caspi et al. (1996) found at least 90% agreement between retrospective reports of activities for a given month on the life history calendar and concurrent reports obtained three years earlier with living arrangements, cohabitation, schooling, employment, and job training.

The high levels of agreement to the month in both of these examples of a life history calendar gain additional impact when one considers that the respondents in both studies were in early adulthood when being surveyed. Specifically, Freedman et al.'s (1988) five-year retrospective report involved 23-year-old respondents reporting on events that had occurred while they were 18 years old; Caspi et al.'s (1996) respondents were 21 years old when they provided retrospective reports of events that happened when they were 18 years old. Young adulthood is marked by a number of transitions in life events, and levels of agreement are likely to have been compromised by this fluidity. However, there cannot be certainty that the event history calendar methodology improves the quality of retrospective reports in comparison to other surveying techniques, as no such direct comparisons have been made. Nevertheless, in comparing indirectly the high levels of agreement reported with the use of event history calendars for retention intervals of several years (Caspi et al., 1996; Freedman et al., 1988) against the lack of reporting with traditional survey questions of events at the same or greater levels of distinctiveness after a one-year retention interval (Cannell et al., 1965; Cash & Moss, 1972), the event history calendar methodology gives the strong impression that its use promotes the ability of respondents to accurately peer further back into their personal pasts.

Potential Applications. For the most part, event history calendars have been used in situations in which respondents were asked to report on major life events (Caspi et al., 1996; Freedman et al., 1988; Furstenberg, Brooks-Gunn, & Morgan, 1987; Kessler & Wethington, 1991; Lyketsos et al., 1994) such as

residence changes, marriages and cohabitation, births and deaths, employment and schooling changes, felony crimes, and periods of serious physical and mental illnesses. In principle, potentially less distinctive events such as health visits to physicians, changes in day-care arrangements for children, and consumer purchases can also be measured using event history calendars. Of course, with mundane events, the length of time used to gather retrospective reports will need to be adjusted to reflect what potentially can be recoverable, and this depends on the degree to which the events of interest lack distinctiveness. For mundane consumer purchases, such as visits to restaurants, a recall period of a few months or so may be reasonable, but not much beyond this.

In addition to recall periods, the units of measurement need to be considered. With very lengthy recall periods, such as a period of many years with life history calendars, using months as the unit of measurement is reasonable. With shorter recall periods, perhaps one or a few years, weeks may adequately serve as units; days can serve as optimal units with even shorter recall periods such as a month or two. In summary, there is considerable flexibility in using event history calendars as the length of recall periods and unit sizes can be adjusted to meet the demands of the kind of events that researchers are interested in measuring.

An important consideration in the use of event history calendars is whether there are multiple timelines, each reflecting an autobiographical theme, that are appropriate to use and which will encourage retrieval of instances of interdependence among these themes. As has been noted, the structure of autobiographical memory can be considered as reflecting many hierarchical layers of extended events that are nested within one another, all adherents within the same theme (e.g. Barsalou, 1988). In addition, there are a number of themes, each with its own hierarchical structure, which also contain points of association with the hierarchies of other themes. Thus, themes will be reflected in various levels of the hierarchy, and one need not use only high-order timelines (e.g. seeking to examine lifetime themes in long-term relationships such as "living with 'X' ") in order to benefit from theme interdependence. Instead, a researcher can restrict penetration to lower-level events that can be reflected in shorter timelines (e.g. "'X's infancy'").

As one example, consider health events. Researchers have often been interested in health utilisation and gaining accurate reports of such events as hospitalisation and visits to physicians' offices. Although not directly identified as an autobiographical theme, the status of one's health certainly can have as much impact in some people's lives as their relationships, living arrangements, and work histories. In terms of extended events, extended periods of illness, injury recuperation, and hospitalisations could adhere to a health status theme, and each could be reflected in separate timelines. Specific events such as surgeries and health visits also are thematically related autobiographical events.

As health visits are often associated with other health events such as hospitalisations, in providing timelines for both types of events respondents can be encouraged to coordinate when each type of event had occurred. In terms of other themes, time off from work, and care of children can be affected by the status of one's health. In constructing an event history calendar, interest in gaining accurate information on rather mundane events such as health visits will indicate the use of recall periods of fairly short duration (a year or two). Nevertheless, there is considerable opportunity to seek multiple cues from the onset and resolution of a number of different extended and specific events in reconstructing health-related events using an event history calendar methodology.

Enhancements to Event History Calendars

There is no doubt that the efficacy of event history calendars can be enhanced by further development. Two particular methods show the greatest potential, using landmarks to aid the dating of past events, and encouraging parallel retrieval. Both methods rely on a more extensive use of temporal and thematic cues that would otherwise be used only spontaneously and erratically by respondents.

Landmarks. Particular landmark events are so noteworthy that respondents are able to locate these events either exactly or nearly exactly in time, and they can be used as anchors to locate when other events happened (Loftus & Marburger, 1983). Means et al. (1989) noted the spontaneous use of landmarks in the dating of health events in a pilot study with volunteer subjects. The authors note (1989, p.25) that landmark events such as "holidays, weddings, parties, job changes, and major health events" were used to date health visits for a 12-month period. There is no doubt that in the use of event history calendars, respondents spontaneously generate their own landmarks. However, recall quality is likely to be enhanced if, in the administration of the calendar, landmarks are systematically introduced. As one example, in the use of a 12-month events history calendar, Kessler and Wethington (1991) used holidays as pre-indicated landmarks and found high reliabilities in event reporting between spouses.

Importantly, the use of landmarks appears not only to assist in event dating, but also to improve the ability to remember what happened, and how often, by assisting in the disentangling of summarised events. As abstractions of repeated events, summarised events are often characterised as being blended representations of specific events, in which any distinctive episodic information that can distinguish one particular episode from another is not retained (Linton, 1982; Means & Loftus, 1991; Means et al., 1989). In this manner, summarised events, although important in that their recall promotes an understanding of the

consistencies that present themselves in our world, are seen as being the bane of survey researchers. For example, Means et al. (1989; Means & Loftus, 1991) noted that recurrent health visits for the same health problem, which are likely to be represented as summarised events, were recalled less well than non-recurrent health visits. However, a study condition that included assisting respondents to generate personally meaningful landmarks along a 12-month calendar led to an increased ability among respondents to remember distinctive episodic information with recurrent visits, improving the accuracy of response.

In assisting respondents to use landmarks, researchers should note that landmarks can be cyclic or linear in nature (Thompson et al., 1996). Cyclic landmarks are those that repeat after certain intervals of time, such as a year. Annual celebrations such as holidays, birthdays, and anniversaries are examples of cyclic landmarks whose dates will often be known. Linear landmarks, on the other hand, are one-time occurrences. Persons may remember such noteworthy dates as when they left for a European vacation, or dates in which residence moves were made. First experiences can also provide linear landmarks, such as the date that one entered a new job, the time that one first drove a car alone, or the occurrence of natural disasters (Pillemer et al., 1988; Robinson, 1992). Birth, deaths, and marriages are linear landmarks, although their later annual remembrances are cyclic landmarks. Finally, researchers may also want to remind respondents about seasonal changes that may be associated with their life routines, such as academic calendars (autumn semester, spring semester, summer) that provide information on distinctive periods of activity (Robinson, 1986).

Whether one encourages the use of cyclic landmarks may depend on the length of recall. Cyclic landmarks are likely to lose their distinctiveness as recall periods include more cycles. *Scale effects* in autobiographical recall indicate that people will at times be better able to locate an event on smaller time scales (e.g. month of occurrence) than on larger time scales (e.g. year of occurrence; see Friedman, 1993). Thus, although a cyclic landmark such as a holiday may help respondents locate in which month an event occurred, confusion may arise as to which year, as the cyclic landmark does not provide any distinctive information concerning the year. Thus, it may be best to focus on linear landmarks for longer recall periods.

Parallel Retrieval. The effectiveness of landmarks largely depends on their effectiveness in conjuring up associated information that resides in other themes or domains of one's autobiographical experience. For example, one may be able to reconstruct that notification of a life-threatening disease came shortly before the celebration of one's child's birthday because the birthday celebration had been somewhat tainted by the news. Similarly, the remembrances of events in one particular theme may conjure up remembrances in other themes, both in terms of what events happened and when. Although respondents are likely to

engage in parallel retrieval spontaneously, the quality of report may be even more enhanced by structuring the interviewing to encourage the use of parallel retrieval.

Structuring the occurrence of parallel retrieval may be accomplished either following or during the completion of the calendar. Encouraging parallel retrieval can be done by engaging the respondent in a calendar review in which the occurrence and timing of events in the various themes can be cross-checked against each other, and modifications to the calendar entries can accompany any additions or changes that the respondent may suggest (Kessler & Wethington, 1991). Parallel retrieval can also be encouraged by directly focusing the respondents' attention on the themes answered first in the calendar while they are working on those themes presented later in the interview (Fricke, Syed, & Smith, 1986). To be most effective in the use of this strategy (and also to encourage spontaneous parallel retrieval), the themes presented first must be ones that are likely to be related to subsequent themes. Freedman et al. (1988) had their first timeline devoted to changes in residence. Not only is this timeline one that is relatively easy for respondents to answer, and one that will assist in motivating the respondent to provide responses for other timelines, but places of residence is also interrelated with other themes such as the relationships that one has had and the work arrangements that exist across the recall period.

As a final thought, given that autobiographical knowledge is dependent on conceptions of the self, researchers may be able to capitalise on these self-conceptions while encouraging parallel retrieval. Providing respondents with labels of the roles that they had assumed during different periods of their lives may serve as cues regarding the various interrelationships among themes. As examples, respondents may be reminded of work-related events that had occurred shortly after becoming a parent, or checks can be made that their becoming a supervisor at work was contemporaneous with becoming a spouse.

CONCLUSION

Recent work on the structure of autobiographical memory suggests that the recall of retrospective reports in surveys can be improved by using the multiple cues and pathways that are available within the autobiographical memory network. Traditional survey questions do not capitalise on these available cues, as they tend to segment the various themes of respondents' pasts. Event history calendars, on the other hand, encourage respondents to appreciate the interrelatedness of various themes which serve to cue memories both within and across these themes. Given the potential of event history calendars for improving the accuracy and completeness of survey report, survey researchers should seriously consider their use.

Direct comparisons in the quality of respondent reports between traditional survey questions and event history calendars are lacking. Indirect com-

parisons, however, do indicate that improvements are likely with event history calendars. In addition, the benefits of calendars ought to be maximised with various enhancements based on what is known to facilitate people's abilities to accurately remember their pasts. However, additional research needs to be conducted that directly compares the quality of responses gained with traditional survey questions and event history calendars, and which also compares the costs of each methodology, considering such aspects as the training of interviewers, interviewing time, and the coding and analyses of data.

Assessing costs is important because reduction of survey errors is almost always associated with some increase in costs (Groves, 1989). The ultimate success of event history calendars will be judged by whether improvements in retrospective reports are shown to be cost-effective in survey administration. For example, exploratory cognitive work has shown that retrospective reports of mundane dietary events can be dramatically improved by using techniques that encourage greater access to cues than is typical in survey questions, but at considerable cost in interviewing time (Fisher & Quigley, 1992). On the other hand, any improvements in accuracy in the use of event history calendars will tend to offset such cost increases, as valid estimates of population characteristics can be obtained with fewer respondents than those needed with traditional survey questions. In summary, the potential degree of improvement in the quality of survey report appears well situated to ensure cost-containment, particularly when the survey topics seek the reporting of those past events that have affected people's lives in several different ways.

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